

Notes: Maggi and Staiger (QJE, 2011)

The Role of Dispute Settlement Procedures in International Trade Agreements

Contents

1	Big picture	2
2	Incremental contribution of the paper	3
3	Institutional background and conceptual framing	3
3.1	Why WTO disputes are not just enforcement disputes	3
3.2	Why contract design and DSB design must be studied together	4
4	The basic static model	4
4.1	Players, policy, and states	4
4.2	First-best policy	4
4.3	No ex post transfers	5
5	Contract language and contract forms	5
5.1	Vague language	5
5.2	Three contract forms	5
6	DSB mandates and possible institutions	6
6.1	Nonactivist and activist roles	6
6.2	Six candidate institutions	6
7	Disputes and DSB accuracy	6
7.1	DSB information	6
7.2	Litigation costs	7
7.3	Timing of the static game	7
8	Static equilibrium under activist DSB roles	7
8.1	Filing condition	7
8.2	Importer protection condition	7
8.3	Key equilibrium logic	7
8.4	Expected loss under gap-filling	8
9	Optimal static institution	8
9.1	Why modification is dominated	8
9.2	Proposition 1: optimal activism and DSB accuracy	8
9.3	Interpretation of Proposition 1	8
9.4	Why DSB use is not a performance metric	9
10	Selection of disputes and pro-trade bias	9
10.1	Observed pro-trade bias in WTO rulings	9
10.2	Bias in DSB rulings	9
10.3	Bias in policy outcomes	9
10.4	Proposition 2: inverse relationship between ruling bias and policy bias	10

11 Dynamic extension: precedent setting	10
11.1 Why precedent matters	10
11.2 Two-period structure	10
11.3 Basic trade-off of precedent	10
11.4 Differential loss under precedent	10
12 Main precedent results	11
12.1 Remark 1: precedent increases first-period disputes	11
12.2 Proposition 3: precedent and DSB accuracy	11
12.3 Proposition 4: precedent and patience	11
13 Tables and figures	12
13.1 Figure I: Bias in DSB Rulings and Policy Outcomes	12
14 What the paper says about WTO design	12
14.1 Interpretation role	12
14.2 Gap-filling role	13
14.3 Modification role	13
14.4 Precedent in WTO disputes	13
15 Identification and empirical interpretation	13
15.1 What is theoretical rather than empirical	13
15.2 How to read the model in relation to real WTO disputes	13
16 Limitations and extensions	13
16.1 No ex post transfers	13
16.2 DSB information is summarized by an error probability	14
16.3 No explicit self-enforcement problem	14
16.4 Domestic courts differ	14
17 Short summary	14
18 Conclusion	15
18.1 Core conclusion	15
18.2 Economic intuition	15
18.3 Incremental contribution revisited	15
18.4 Takeaway	15

1 Big picture

- **Question.** What role should dispute settlement procedures play in international trade agreements? In particular, should the dispute settlement body merely enforce clear obligations, or should it actively complete incomplete contracts by interpreting vague language, filling contractual gaps, or granting exceptions to rigid rules?
- **Institutional motivation.** Standard economic models often treat disputes as enforcement problems. The paper argues that many WTO disputes are instead about contract incompleteness: provisions are vague, the agreement is silent, or rigid obligations may seem inappropriate in an unforeseen state.
- **Core idea.** A trade agreement and the mandate of the Dispute Settlement Body (DSB) should be designed jointly. Contract form and court mandate are complements: the value of a vague contract depends on whether the DSB may interpret it, and the value of leaving discretion depends on whether the DSB may fill gaps.
- **Three forms of contractual incompleteness.** The paper distinguishes rigidity, discretion, and vagueness. Rigidity means the contract imposes a fixed obligation even when exceptions may be efficient. Discretion means the agreement is silent, leaving policy choice to the importer. Vagueness means the contract uses ambiguous language that is clear in some states but undefined in gray-area states.
- **Three activist DSB roles.** The DSB may interpret vague provisions, fill gaps where the contract is silent, or modify rigid obligations by granting exceptions. The paper compares these activist roles with a nonactivist DSB that only enforces crisp obligations.
- **Main static result.** The optimal degree of DSB activism depends on DSB accuracy. If the DSB is highly accurate, the best institution is a silent contract with a gap-filling DSB. If DSB accuracy is intermediate, the best institution is a vague contract with an interpretive DSB. If the DSB is inaccurate, the best institution uses a nonactivist DSB with either a vague or rigid contract.
- **Modification result.** The paper finds no role for a modification mandate in the optimal institution. If the DSB is going to decide whether protection should be allowed, it is better to leave a gap and let the DSB fill it than to write a rigid free-trade rule and let the DSB modify it ex post.
- **Off-equilibrium value of courts.** The DSB can improve efficiency even when it is rarely used. If governments expect a sufficiently accurate DSB ruling, the threat of litigation can discipline behavior and induce efficient policies without actual disputes.
- **Disputes are not necessarily good news.** In equilibrium, disputes tend to arise when one party tries to exploit contract incompleteness and the DSB may make an error. Therefore, frequent DSB use is not a reliable measure of institutional success.
- **Precedent result.** In a dynamic model, precedent saves future litigation costs by clarifying rights and obligations, but it also raises the stakes of early litigation and therefore increases first-period disputes. Precedent is more likely to be useful when DSB information is less accurate and when governments discount the future more heavily.

2 Incremental contribution of the paper

- **From enforcement to interpretation.** The paper moves beyond the standard view that trade disputes are mainly enforcement problems. It models disputes as responses to incomplete contracts and asks what institutional design should accompany different forms of incompleteness.
- **Joint design of contract and court.** The paper’s central methodological contribution is to treat the trade agreement and the DSB mandate as one institutional design problem. The optimal contract cannot be evaluated separately from the court’s authority.
- **Formal treatment of vague trade-contract language.** The paper introduces a simple formal model of vague language using a three-valued truth function: a provision may be true, false, or undefined. This captures the fact that phrases such as “substantial injury” are clear in extreme cases but ambiguous in intermediate states.
- **Mapping between incompleteness and DSB roles.** The paper connects three incomplete-contract forms to three DSB mandates: discretion to gap-filling, vagueness to interpretation, and rigidity to modification.
- **A ranking of optimal activism.** The model provides a clean comparative-static result: as DSB accuracy rises, the optimal DSB role becomes more activist. This gives a formal logic for when courts should be restrained and when they should complete the contract.
- **Selection explanation for pro-trade rulings.** The paper explains why complainants often win WTO cases. A pro-trade bias in observed DSB rulings can arise from selection into disputes, especially when the exporter’s cost of filing is high relative to the importer’s cost of defending.
- **Ruling bias versus policy bias.** The paper shows that a pro-trade bias in DSB rulings can coexist with an anti-trade bias in equilibrium policy outcomes. The two biases can move in opposite directions because observed rulings are selected from nonrandom disputes.
- **New cost of precedent.** The paper highlights a harmful effect of precedent that is especially relevant for international agreements: precedent raises the future stakes of first-period litigation and can increase early disputes.
- **Institutional relevance.** The framework speaks directly to WTO debates over judicial activism, Article 3.2 of the Dispute Settlement Understanding, and whether WTO rulings should operate as precedent.

3 Institutional background and conceptual framing

3.1 Why WTO disputes are not just enforcement disputes

The paper begins from the observation that many WTO disputes are not simply about whether a country violated an unambiguous promise. Instead, disputes often arise because the agreement contains language that must be interpreted or because the agreement does not specify what should happen in a particular circumstance. In legal terms, the agreement may be incomplete through gaps, ambiguity, or rigid provisions that do not fit the realized state.

This distinction matters because the appropriate role of the DSB differs across cases. If the contract is clear, the DSB can enforce. If the contract is vague, the DSB may interpret. If the contract is silent, the DSB may fill a gap. If the contract is rigid, the DSB may be asked to grant

an exception. The paper asks which of these roles should be deliberately assigned to the DSB ex ante.

3.2 Why contract design and DSB design must be studied together

A vague clause is useful only if its ambiguity can be resolved in a sensible way when disputes arise. A silent contract may be efficient only if the DSB can fill gaps or if private incentives discipline behavior without disputes. A rigid obligation can be too strong if efficient policy depends on the state of the world. Therefore, the contract form and DSB mandate are jointly chosen.

The paper's institutional design object is therefore:

$$\text{Institution} = \text{Contract form} + \text{DSB mandate}.$$

The governments choose this institution ex ante to maximize expected joint surplus.

4 The basic static model

4.1 Players, policy, and states

There are two governments: an importing government and an exporting government. The importing government chooses a binary trade policy in one industry:

$$\tau \in \{\text{FT}, \text{P}\},$$

where FT denotes free trade and P denotes protection. The state of the world is

$$s = (s_1, s_2, \dots, s_N),$$

where each s_i is a binary event. For example, the state may summarize whether there is an import surge, whether the domestic industry shuts down, or whether workers are unemployed.

The importing government's payoff is $\omega(\tau; s)$, and the exporting government's payoff is $\omega^*(\tau; s)$. Define the importer's gain from protection as

$$\gamma(s) = \omega(\text{P}; s) - \omega(\text{FT}; s),$$

and the exporter's gain from protection as

$$\gamma^*(s) = \omega^*(\text{P}; s) - \omega^*(\text{FT}; s).$$

The paper assumes

$$\gamma(s) > 0 \quad \text{and} \quad \gamma^*(s) < 0$$

for all states. Thus, the importing government privately prefers protection, while the exporting government dislikes protection.

4.2 First-best policy

Joint surplus is

$$\Omega(\tau; s) = \omega(\tau; s) + \omega^*(\tau; s).$$

Define the joint gain from protection as

$$\Gamma(s) = \gamma(s) + \gamma^*(s) = \Omega(\text{P}; s) - \Omega(\text{FT}; s).$$

The first-best policy is protection if $\Gamma(s) > 0$ and free trade if $\Gamma(s) \leq 0$. Define

$$\sigma_P = \{s : \Gamma(s) > 0\}, \quad \sigma_{FT} = \{s : \Gamma(s) \leq 0\}.$$

The governments know $\Gamma(s)$, but the DSB does not. This unverifiability of payoffs is what prevents the contract from simply writing the first-best policy as a complete contingent contract.

4.3 No ex post transfers

The model rules out ex post transfers between governments after the state is realized. This is important because, without transfers, the governments cannot renegotiate to achieve the first best once their interests conflict. It also reflects the fact that WTO dispute resolution rarely uses direct monetary transfers.

5 Contract language and contract forms

5.1 Vague language

A central contribution of the paper is its formalization of vague language. The contract may use a sentence ν , such as “there is substantial injury to the domestic industry due to increased imports.” The sentence has a three-valued truth function:

$$\nu(s) = \begin{cases} \text{True,} & s \in T, \\ \text{False,} & s \in F, \\ \text{Undefined,} & s \notin T \cup F. \end{cases}$$

Here T is the set of extreme states where the sentence is clearly true, F is the set of states where it is clearly false, and the remaining states form a gray area where the provision is ambiguous.

The assumptions are

$$T \subset \sigma_P, \quad F \subset \sigma_{FT}.$$

Thus, when the vague sentence is clearly true, protection is efficient; when it is clearly false, free trade is efficient. Ambiguity arises only in intermediate states.

5.2 Three contract forms

The paper compares three stylized contract forms.

1. **Silent contract** (D). The contract is empty or silent. It leaves full discretion to the importer.
2. **Rigid contract** (R). The contract imposes free trade in all states: protection is never allowed.
3. **Vague contract** (V). The contract says protection is allowed if and only if ν . This creates clear rights in T , clear obligations in F , and ambiguity outside $T \cup F$.

These contract forms correspond to three sources of incompleteness:

- **Discretion:** the contract is silent.
- **Rigidity:** the contract imposes a fixed rule.
- **Vagueness:** the contract uses ambiguous language.

6 DSB mandates and possible institutions

6.1 Nonactivist and activist roles

The DSB always enforces crisp obligations in the background. The key question is whether it can go further. The possible mandates are:

- **Nonactivist mandate** (n): the DSB only enforces clear obligations.
- **Interpretation mandate** (i): the DSB interprets ambiguous provisions.
- **Gap-filling mandate** (g): the DSB fills gaps where the contract is silent.
- **Modification mandate** (m): the DSB grants exceptions to rigid obligations.

The interpretation mandate is intermediate activism; gap-filling and modification are more activist because they allow the DSB to add to or diminish rights and obligations.

6.2 Six candidate institutions

The candidate institutions are summarized as follows.

Candidate institutions: contract form and DSB mandate.

DSB role	Silent contract	Rigid contract	Vague contract
Nonactivist	D_n	R_n	V_n
Activist	D_g : fills gaps	R_m : allows exceptions	V_i : interprets

The silent nonactivist institution D_n essentially delivers the noncooperative outcome because the importer is unconstrained and prefers protection. The rigid nonactivist institution R_n prevents protection even when protection would be jointly efficient. The vague nonactivist institution V_n constrains the importer in clear free-trade states but leaves discretion in ambiguous gray-area states. The activist institutions potentially improve on these outcomes, but they also create litigation costs and DSB-error risk.

7 Disputes and DSB accuracy

7.1 DSB information

The DSB observes the state s , but it does not observe $\Gamma(s)$, the joint gain from protection. If invoked, it receives a noisy signal and issues a ruling. Let

$$q(s)$$

be the probability that the DSB issues the wrong ruling in state s . The paper parameterizes DSB noise as

$$q(s) = q \cdot k(s),$$

where $q \in [0, 1]$ is an inverse measure of overall DSB accuracy. A lower q means a more accurate DSB.

7.2 Litigation costs

If the exporter files a complaint, the exporter incurs cost $c^* > 0$, and the importer incurs cost $c > 0$. These costs capture legal, administrative, and burden-of-proof costs. Total litigation cost is

$$c^{all} = c + c^*.$$

7.3 Timing of the static game

The static game is:

1. The institution is chosen.
2. The state s is realized.
3. The importer chooses $\tau \in \{\text{FT}, \text{P}\}$.
4. The exporter decides whether to file a complaint.
5. If invoked, the DSB issues a ruling.
6. Payoffs are realized.

8 Static equilibrium under activist DSB roles

8.1 Filing condition

Suppose the importer chooses protection. The exporter files if the expected benefit exceeds the filing cost:

$$\Pr(\text{DSB ruling is FT} \mid s) \cdot |\gamma^*(s)| > c^*.$$

This is the exporter's filing condition.

8.2 Importer protection condition

If filing would occur, the importer still chooses protection if its expected benefit exceeds the litigation cost:

$$\Pr(\text{DSB ruling is P} \mid s) \cdot \gamma(s) > c.$$

8.3 Key equilibrium logic

Under the gap-filling institution D_g , the equilibrium has the following structure:

- If $s \in \sigma_{FT}$, free trade is efficient. The importer may still choose protection if DSB error is sufficiently likely. The exporter then files. Thus, disputes in these states are driven by the importer trying to get away with inefficient protection.
- If $s \in \sigma_P$, protection is efficient. The exporter may still file if DSB error is sufficiently likely. Thus, disputes in these states are driven by the exporter trying to force inefficient free trade.

This logic implies two important messages:

1. DSB use is more likely when the DSB is less accurate.
2. The DSB improves efficiency mainly through off-equilibrium threats, not through frequent actual rulings.

8.4 Expected loss under gap-filling

Let $\hat{\sigma}_{FT}$ be the set of states where free trade is efficient but the importer chooses protection and the exporter files. Let $\hat{\sigma}_P$ be the set of states where protection is efficient but the exporter files. The expected loss under D_g , relative to the first best, is

$$L(D_g) = \sum_{s \in \hat{\sigma}_{FT} \cup \hat{\sigma}_P} p(s) \left[qk(s)|\Gamma(s)| + c^{all} \right].$$

The loss has two parts: the expected cost of DSB error and the litigation cost. When the DSB is sufficiently accurate, neither party wants to trigger litigation in equilibrium, and $L(D_g) = 0$. The first best can then be achieved even though the contract is silent and the DSB is imperfect.

9 Optimal static institution

9.1 Why modification is dominated

The institution R_m imposes a rigid free-trade rule but allows the DSB to grant exceptions. If the DSB is invoked, it effectively decides whether protection should be allowed. This gives the same policy outcome as leaving the contract silent and letting the DSB fill the gap under D_g . But the rigid contract entails a writing cost, while the silent contract is costless. Therefore, R_m is dominated by D_g .

The intuition is simple: if the DSB is trusted enough to decide when exceptions are appropriate, it is better to leave a gap than to write a rigid rule and later ask the DSB to undo it.

9.2 Proposition 1: optimal activism and DSB accuracy

There exist thresholds q_1 and q_2 , with

$$0 < q_1 \leq q_2 \leq 1,$$

such that:

1. If $q < q_1$, the optimal institution is D_g : a silent contract with a gap-filling DSB.
2. If $q_1 < q < q_2$, the optimal institution is V_i : a vague contract with an interpretive DSB.
3. If $q > q_2$, the optimal institution has a nonactivist DSB and uses either V_n or R_n .

9.3 Interpretation of Proposition 1

The optimal DSB mandate becomes more activist as DSB information improves. When the DSB is highly accurate, governments can leave broad discretion and rely on the DSB's off-equilibrium threat to discipline opportunism. When DSB accuracy is intermediate, a vague contract is useful because it gives the DSB a narrower domain: the DSB interprets only gray-area states, while clear states are handled by the contract itself. When the DSB is inaccurate, activism is too dangerous, and the institution should rely on nonactivist enforcement of clear obligations.

9.4 Why DSB use is not a performance metric

The model implies a nonmonotonic relation between DSB use and institutional performance. A highly accurate activist DSB may be used rarely because its threat is enough to induce efficient behavior. A less accurate activist DSB may be used more frequently precisely because governments try to exploit its possible errors. Therefore, a high dispute frequency does not necessarily mean that the institution is functioning well, and a low dispute frequency does not necessarily mean that the DSB is irrelevant.

10 Selection of disputes and pro-trade bias

10.1 Observed pro-trade bias in WTO rulings

The paper addresses the empirical fact that complainants have often won GATT/WTO disputes. This looks like a pro-trade bias because complainants are usually exporters challenging import barriers. The model shows that such a bias can emerge from selection into litigation, even if the DSB itself is unbiased *ex ante*.

10.2 Bias in DSB rulings

Suppose disputes would be unbiased if they were randomly initiated. In equilibrium, however, disputes are selected. The probability that the DSB rules for free trade conditional on a filing is

$$\Pr(\tau^{DSB} = \text{FT} \mid \text{file}) = \frac{\sum_{s \in \hat{\sigma}_{FT}} (1 - q(s))p(s) + \sum_{s \in \hat{\sigma}_P} q(s)p(s)}{\sum_{s \in \hat{\sigma}_P} p(s) + \sum_{s \in \hat{\sigma}_{FT}} p(s)}.$$

There is a pro-trade bias in rulings if this probability exceeds 1/2.

The key comparative static is:

- If the exporter's filing cost c^* is high relative to the importer's dispute cost c , the exporter files mainly when the importer is trying to get away with inefficient protection.
- Therefore, observed disputes are more likely to be cases where free trade is efficient, so the DSB often rules for free trade.

10.3 Bias in policy outcomes

The bias in observed DSB rulings is not the same as the bias in equilibrium policy outcomes. Policy outcomes exhibit a pro-trade bias if

$$\Pr(\hat{\tau}(D_g) = \text{FT}) > \frac{1}{2},$$

where $\hat{\tau}(D_g)$ is the equilibrium policy under D_g . The paper shows that policy outcomes tend to be pro-trade when the exporter's filing cost is low relative to the importer's cost.

Thus, the same conditions that generate a pro-trade bias in rulings tend to generate an anti-trade bias in actual policy. If exporters file only in very strong cases, observed rulings look pro-trade; but many inefficient protection choices may go unchallenged.

10.4 Proposition 2: inverse relationship between ruling bias and policy bias

For the activist institutions D_g and V_i :

1. There is a pro-trade bias in DSB rulings if and only if $c^* > y(c)$, where $y(\cdot)$ is weakly increasing.
2. There is a pro-trade bias in policy outcomes if and only if $c^* < z(c)$, where $z(\cdot)$ is weakly increasing.
3. In the symmetric case, the condition for pro-trade ruling bias is exactly the condition for anti-trade policy bias.

11 Dynamic extension: precedent setting

11.1 Why precedent matters

The static model asks how activist the DSB should be. The dynamic model asks whether DSB rulings should set precedent for future disputes. A ruling sets precedent if, after the DSB rules in state s , the contract is completed for that state in future periods. If the same state occurs again, the earlier ruling determines the rule.

11.2 Two-period structure

The dynamic model has two periods after the institution is designed. The state is independent and identically distributed across periods, and governments discount period 2 by δ . If the DSB has precedent-setting authority, a period-1 ruling for state s binds the institution in period 2 if state s recurs.

11.3 Basic trade-off of precedent

Precedent has two opposing effects.

- **Benefit: future litigation-cost savings.** Once a state has been ruled on, uncertainty about rights and obligations in that state is removed. If the state recurs, there is no need to litigate again.
- **Cost: more first-period disputes.** Because a first-period ruling affects future obligations, both governments have stronger incentives to litigate in period 1. The importer may try harder to obtain a favorable precedent allowing protection; the exporter may try harder to obtain a favorable precedent requiring free trade.

The second effect is the paper's novel contribution to the precedent literature. It is especially important in international settings because the same governments interact repeatedly before the same legal system.

11.4 Differential loss under precedent

Let D_g^P denote the gap-filling institution with precedent. The loss difference between precedent and no precedent is

$$L(D_g^P) - L(D_g) = \sum_{s \in \hat{\Delta}} p(s)[qk(s)|\Gamma(s)| + c^{all}] + \delta \sum_{s \in \hat{\Delta}} p^2(s)qk(s)|\Gamma(s)| - \delta \sum_{s \in \hat{\sigma}} p^2(s)c^{all}.$$

The first term is the period-1 cost from additional disputes induced by precedent. The second term is the future cost of erroneous precedents. The third term is the future litigation-cost saving from not relitigating states that would otherwise generate disputes.

12 Main precedent results

12.1 Remark 1: precedent increases first-period disputes

If the DSB plays an activist role, the frequency of first-period disputes weakly rises when precedent is introduced. The reason is that precedent raises the future stakes of litigation. A favorable first-period ruling not only affects the current dispute but also determines the rule if the same state recurs.

12.2 Proposition 3: precedent and DSB accuracy

For a given activist DSB role, as DSB noise q increases from 0:

1. Initially, precedent has no effect.
2. Then precedent becomes strictly undesirable.
3. Finally, as q approaches 1, precedent becomes strictly desirable.

The intuition is subtle. When the DSB is very accurate, there may be no disputes even without precedent; introducing precedent may create additional first-period disputes without saving much future litigation. When the DSB is very inaccurate, disputes already occur in many states, so precedent mainly saves future litigation costs. Therefore, precedent is more likely to be useful when DSB accuracy is low.

12.3 Proposition 4: precedent and patience

For an intermediate range of DSB noise, precedent is desirable when δ is sufficiently low but undesirable when δ is sufficiently high.

This result follows because a high δ magnifies the future stakes of first-period litigation, increasing the incentive to create precedent. If governments are patient and expect to interact repeatedly, precedent can create too much strategic litigation early on. If governments are impatient, precedent does not greatly distort first-period incentives, and future litigation-cost savings can dominate.

13 Tables and figures

13.1 Figure I: Bias in DSB Rulings and Policy Outcomes

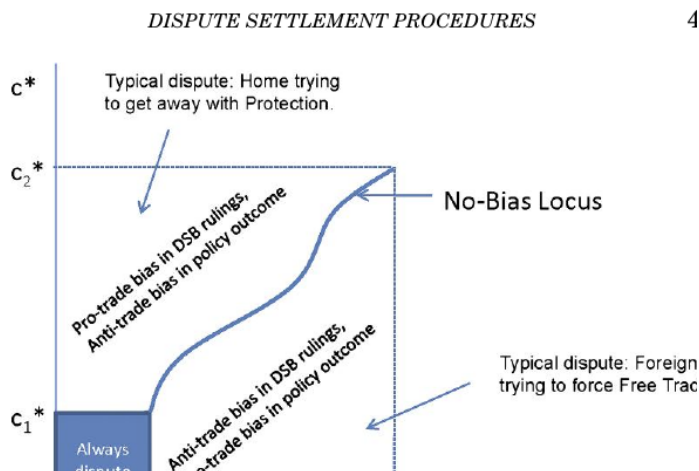


Figure I: Bias in DSB Rulings and Policy Outcomes.

Read:

- The horizontal axis is the importer's litigation cost c , and the vertical axis is the exporter's litigation cost c^* .
- The no-bias locus separates regions where the typical dispute is driven by the home/importing country trying to get away with protection from regions where the foreign/exporting country is trying to force free trade.
- Above the no-bias locus, c^* is relatively high. Exporters file only in stronger cases. Observed DSB rulings therefore look pro-trade, but equilibrium policy outcomes are biased against trade because some inefficient protection goes unchallenged.
- Below the no-bias locus, c^* is relatively low. Exporters are more willing to challenge protection, including cases where protection may actually be efficient. Observed rulings then look less pro-trade or anti-trade, while equilibrium policy outcomes can be biased toward free trade.
- The shaded lower-left region indicates that when both parties' dispute costs are sufficiently low, disputes always occur.

Economic message: The figure visualizes the key selection logic. Observed rulings are not a random sample of states; they are selected by filing incentives. Therefore, a pro-trade bias in DSB rulings can coexist with an anti-trade bias in actual policy outcomes.

14 What the paper says about WTO design

14.1 Interpretation role

The WTO Dispute Settlement Understanding explicitly says that the dispute settlement system clarifies existing provisions and cannot add to or diminish rights and obligations. This language is closest to an interpretation mandate rather than a gap-filling or modification mandate. The paper's

model provides a reason why interpretation may be optimal when DSB accuracy is intermediate: vague language can economize on writing costs, while limiting the DSB's domain to ambiguous states.

14.2 Gap-filling role

The model says gap-filling can be optimal if the DSB is highly accurate. In that case, leaving gaps is not necessarily a failure of design. A gap may be efficient because the threat of an accurate DSB ruling disciplines governments without requiring a highly detailed ex ante contract.

14.3 Modification role

The model is skeptical of a modification role. If the institution needs flexibility, it is better to leave discretion or vagueness ex ante than to write a rigid obligation and give the DSB power to undo it. This is why R_m is dominated by D_g in the model.

14.4 Precedent in WTO disputes

The paper interprets precedent as an institutional design choice. Precedent can create predictability and save future litigation costs, but it also makes early cases more valuable and therefore more heavily contested. The implication is not that precedent is always bad, but that its value depends on DSB accuracy, governments' patience, and the repeated nature of interaction.

15 Identification and empirical interpretation

15.1 What is theoretical rather than empirical

The paper is a formal theory paper. It does not estimate the model structurally. Its empirical content lies in comparative statics and institutional interpretations:

- DSB activism should be higher when courts are better informed.
- More frequent disputes do not necessarily indicate a more successful institution.
- Observed pro-complainant or pro-trade rulings may reflect selection, not court bias.
- Precedent should be less attractive in settings with repeated interactions among the same litigants.

15.2 How to read the model in relation to real WTO disputes

The model is intentionally stylized. It uses one policy, two countries, binary protection, no ex post transfers, and exogenous DSB accuracy. These simplifications allow the paper to isolate the design logic. The model is not meant to reproduce all details of WTO litigation, but to clarify why contract incompleteness, court activism, and precedent interact in nontrivial ways.

16 Limitations and extensions

16.1 No ex post transfers

The assumption of no ex post transfers fits international trade agreements better than domestic contracts, but it is still stark. If transfers were available and cheap, parties could renegotiate around

inefficient rulings or incomplete contract terms. The paper argues that WTO disputes rarely involve direct transfers, so the assumption is reasonable for the setting.

16.2 DSB information is summarized by an error probability

The DSB's signal is reduced to the probability of a wrong ruling. This is useful for deriving clean results, but it abstracts from how hearings, evidence, legal arguments, or expert reports shape DSB beliefs. The authors note that the signal can be interpreted as incorporating information extracted from hearings.

16.3 No explicit self-enforcement problem

The paper assumes DSB rulings and crisp obligations are enforced. The authors discuss how the logic might extend to self-enforcing agreements if the DSB helps transmit information to third countries that can support enforcement through retaliation or reputational mechanisms.

16.4 Domestic courts differ

The authors caution against directly applying the model to domestic legal systems. Domestic systems often have monetary remedies and many one-shot litigants, while trade agreements involve repeated interactions among governments. This makes the strategic-litigation cost of precedent more important in international law than in many domestic settings.

17 Short summary

- The paper studies how dispute settlement procedures should be designed when trade agreements are incomplete.
- It distinguishes three forms of incompleteness: discretion, rigidity, and vagueness.
- It compares nonactivist enforcement with three activist DSB roles: gap-filling, interpretation, and modification.
- The optimal DSB mandate depends on DSB accuracy: high accuracy supports gap-filling, intermediate accuracy supports interpretation, and low accuracy supports nonactivism.
- A modification mandate is never optimal because a silent contract with gap-filling dominates a rigid contract with ex post modification.
- The DSB can improve outcomes even when no disputes occur because its off-equilibrium threat disciplines policy choices.
- Equilibrium disputes often reflect inefficient attempts to exploit DSB error, so dispute frequency is not a reliable measure of institutional performance.
- Selection into litigation can explain why complainants often win WTO disputes.
- A pro-trade bias in DSB rulings can coincide with an anti-trade bias in policy outcomes.
- Precedent saves future litigation costs but increases first-period disputes by raising future stakes.
- Precedent is more attractive when DSB information is poor and when governments are less patient.

18 Conclusion

18.1 Core conclusion

Maggi and Staiger show that dispute settlement procedures are not merely enforcement devices. They are part of the design of incomplete international trade contracts. The DSB can help complete the agreement by interpreting vague provisions or filling gaps, but activism is valuable only when the DSB is sufficiently well informed.

18.2 Economic intuition

The key economic force is the trade-off between contractual flexibility and judicial error. A complete state-contingent contract would be too costly to write. Incomplete contracts save writing costs but create scope for opportunism. An activist DSB can discipline opportunism, but only if its expected rulings are accurate enough. When DSB accuracy is low, court activism becomes another source of distortion.

18.3 Incremental contribution revisited

The paper's contribution is not just to say that courts matter. It identifies exactly how the optimal court mandate depends on the form of the contract and on court accuracy. It also shows that observed litigation outcomes are selected and can mislead observers about the underlying policy bias or institutional success.

18.4 Takeaway

The most useful reading of the paper is: design the contract and the court together. If the DSB is accurate, leave more discretion and let the court fill gaps. If the DSB is moderately accurate, use vague language and allow interpretation. If the DSB is inaccurate, constrain the court and rely on clearer rules. Be cautious with precedent because it improves future predictability but can create excessive strategic litigation today.